

Ticket 10A Study Report: Rannacher Smoothing Comparison Gate

SURF2026 American Option Risk Surfaces

Codex-assisted implementation report

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Abstract

Ticket 10A adds a controlled Rannacher smoothing comparison after the Ticket 10 Delta and Gamma diagnostics. It does not replace the validated Ticket 04 American CN/PSOR baseline. Instead, it creates a named Rannacher variant that uses two backward-Euler half-substeps over the first ordinary time interval, then resumes ordinary CN/PSOR. The experiment compares baseline and smoothed results for price levels, PSOR convergence, threshold-based boundary curves, and finite-difference Gamma diagnostics. The gate conclusion for the representative American put and dividend-paying American call cases is: keep the baseline for now. The smoothed variant changed prices and boundaries only slightly, but it did not improve the headline strict-mask Gamma metrics in this run.

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1 Role of Ticket 10A in the Whole Solver Project

Tickets 01 through 10 built a validation ladder before this comparison gate:

- Ticket 01 created payoff and Black–Scholes closed-form utilities.
- Tickets 02 and 03 built the grid, spatial operator, and European CN validation mode.
- Ticket 04 implemented the American CN/PSOR obstacle core.
- Tickets 05 through 07 validated American put, no-dividend American call, and dividend-paying American call behavior.
- Ticket 08 added obstacle and complementarity diagnostics.
- Ticket 09 extracted approximate free-boundary curves from $U - \Phi$.
- Ticket 10 computed finite-difference Delta and Gamma diagnostics and warned that Gamma is fragile near payoff kinks, maturity, and extracted boundary regions.

Ticket 10A asks whether a Rannacher-style startup damping step improves those fragile Greek diagnostics enough to justify using it later. It is a comparison gate, not a new default solver. The baseline solver in `src/american_risk_surfaces/solvers/cn_psor.py` remains unchanged.

2 Theory and Methodology Connection

The documented source basis for this ticket is internal:

- the literature map identifies Rannacher smoothing as a classical finite-difference stabilization idea near payoff kinks;
- the formulation note explains American option values, payoff obstacles, and Greeks;
- the solver validation plan requires an explicit unsmoothed-versus-smoothed comparison before smoothing is adopted;
- Ticket 09 and Ticket 10 provide boundary and Greek diagnostic tools used by this comparison.

No BibTeX file is created. The report does not invent external bibliographic details. It uses the project documents as the source basis and treats Rannacher smoothing as a numerical time-stepping device, not an economic model change.

3 Rannacher Smoothing in Plain Language

Crank–Nicolson is accurate and stable for smooth solutions, but option payoffs are not smooth at the strike. At maturity the value equals payoff, so the first time steps start from a kinked function. Gamma is a second derivative, so it can amplify that kink into noisy or exaggerated curvature.

Rannacher smoothing handles this by beginning with a few backward-Euler substeps. Backward Euler is more damping than Crank–Nicolson. The idea is:

1. begin at maturity payoff;
2. take a small number of implicit backward-Euler steps to damp the kink;
3. continue with ordinary Crank–Nicolson after the startup interval.

This should not be used just to make a plot look smoother. It must be checked against prices, boundaries, and solver convergence.

4 Why This Comes After Greek Diagnostics

Ticket 10 first measured the baseline Delta and Gamma behavior. That made the problem visible: Gamma can be fragile near maturity, payoff kinks, and extracted boundaries. Ticket 10A then tests one targeted numerical remedy. This order matters. Without Ticket 10's diagnostics, smoothing could become a vague aesthetic preference. With Ticket 10's diagnostics, smoothing can be judged using evidence.

5 Backward-Euler Substeps Versus Crank–Nicolson

The ordinary American CN/PSOR step solves a linear complementarity problem based on

$$A_{\text{CN}} = I - \frac{1}{2}\Delta\tau L_h,$$

with a right-hand side containing

$$\left(I + \frac{1}{2}\Delta\tau L_h\right) U^n.$$

The Ticket 10A startup substep instead uses backward Euler:

$$A_{\text{BE}} = I - \delta\tau L_h.$$

For each substep, the interior LCP is

$$A_{\text{BE}}u \geq b, \quad u \geq \Phi, \quad (u - \Phi)^\top (A_{\text{BE}}u - b) = 0,$$

where b is the previous value vector adjusted for the new-time boundary values. Ticket 10A's default is two equal substeps over the first ordinary interval:

$$\delta\tau = \frac{\Delta\tau}{2}.$$

6 Implementation Design Decisions

Decision	Reason
<code>rannacher.py</code> is separate	Prevents silent changes to the validated Ticket 04 baseline solver.
Two half-substeps	Matches the planned Rannacher gate and keeps the comparison simple.
<code>rannacher_substeps=0</code> delegates to baseline	Provides an explicit off switch and a testable equivalence path.
Reuse Ticket 04 PSOR	Keeps projection logic identical to the American obstacle core.
Reuse Ticket 09 and 10 diagnostics	Boundary and Gamma comparisons use existing validated diagnostic layers.

Decision	Reason
Gate recommendation is conservative	Smoothing is adopted only if it improves Greeks without material price or boundary distortion.

7 Code-Level Function Explanations

Function or class	Purpose
<code>RannacherSmoothingMetadata</code>	Records whether smoothing is enabled, number and size of substeps, separated PSOR metadata, convergence status, and obstacle violation.
<code>RannacherCNPSORResult</code>	Wraps a standard <code>AmericanCNPSORResult</code> with Rannacher-specific metadata.
<code>backward_euler_psor_step</code>	Solves one implicit-Euler American LCP using the existing PSOR projection solver.
<code>rannacher_crank_nicolson_psor_price</code>	Runs two backward-Euler half-substeps, stores the first main time row, then continues with ordinary CN/PSOR.
<code>experiments/08_rannacher_smoothing_comparison.py</code>	Runs baseline and smoothed representative cases, writes CSV comparisons, and creates figures.

8 Comparison Cases and Parameters

Case	Type	Parameters
<code>american_put_medium</code>	put	$K = 1.0, T = 1.0, r = 0.05, q = 0.02, \sigma = 0.20, S_{\max} = 4.0, M = N = 120.$
<code>dividend_call_medium</code>	call	$K = 1.0, T = 1.0, r = 0.05, q = 0.08, \sigma = 0.20, S_{\max} = 4.0, M = N = 120.$

Scheme setting	Value
Ordinary time interval	$\Delta\tau = 1/120 = 0.0083333333333333.$
Rannacher substeps	2.
Substep size	0.0041666666666667.
Post-startup method	Ordinary CN/PSOR for the remaining main time intervals.
Baseline behavior	Unchanged Ticket 04 American CN/PSOR solver.

9 Price Comparison Tables and Interpretation

Case	Max abs price diff	RMSE diff	Max-diff spot	Gate
american_put_medium	$1.66937904909 \times 10^{-6}$	$4.19362587648 \times 10^{-7}$	1.000000	KEEP_BASELINE_ FOR_NOW
dividend_call_medium	$1.72323866424 \times 10^{-6}$	$4.26128419836 \times 10^{-7}$	1.033333	KEEP_BASELINE_ FOR_NOW

The price differences are very small in both representative cases. This means the Rannacher startup does not materially distort the price level in this run. However, price similarity alone does not justify adopting smoothing. The Greek and boundary comparisons must also support the change.

Case	S/K	Payoff	Baseline	Rannacher	Difference
american_put_medium	0.8	0.200000	0.200381130316	0.200380822124	-3.0819×10^{-7}
american_put_medium	1.0	0.000000	0.066301330539	0.066299661160	-1.6694×10^{-6}
american_put_medium	1.2	0.000000	0.015983679585	0.015982898743	-7.8084×10^{-7}
dividend_call_medium	0.8	0.000000	0.008367161513	0.008366693230	-4.6828×10^{-7}
dividend_call_medium	1.0	0.000000	0.065131176799	0.065129465900	-1.7109×10^{-6}
dividend_call_medium	1.2	0.200000	0.203695017872	0.203694325038	-6.9283×10^{-7}

10 Boundary Comparison Tables and Interpretation

Case	Matched rows	Max shift	Mean shift	Status
american_put_medium	120	0.000540799953	$1.09039294769 \times 10^{-5}$	COMPARABLE
dividend_call_medium	120	0.000663055353	$1.48912617221 \times 10^{-5}$	COMPARABLE

The threshold-based boundary curves remain close. This is reassuring, but it is not proof of exact free-boundary accuracy. Boundary extraction inherits the Ticket 09 threshold and grid uncertainty. The important gate point is that smoothing did not create an obvious boundary distortion in these cases.

11 Greek Comparison Tables and Interpretation

Case	Baseline full max $ \Gamma $	Rannacher full max $ \Gamma $	Baseline strict max $ \Gamma $	Strict change
american_put_medium	30.000000	30.000000	3.180018866	+0.000266535
dividend_call_medium	30.000000	30.000000	2.679887372	+0.000341023

The full-grid maximum $|\Gamma|$ is unchanged because the strongest values remain at fragile regions such as maturity or kink bands. The strict-mask Gamma metric is slightly higher after smoothing in both cases. This does not mean Rannacher smoothing is bad in general. It means this particular comparison does not provide enough evidence to adopt it as the default or even as the preferred Greek-diagnostic path.

12 PSOR Convergence Comparison

Both baseline and Rannacher runs converged for both representative cases. The Rannacher variant records the startup backward-Euler PSOR solves separately from the later CN/PSOR solves. This is important because a smoothed variant would be risky if it improved Gamma while causing hidden non-convergence. That did not occur here.

Case	Baseline converged	Rannacher converged	Obstacle violation
american_put_medium	True	True	0 for both methods
dividend_call_medium	True	True	0 for both methods

13 Figure Interpretation

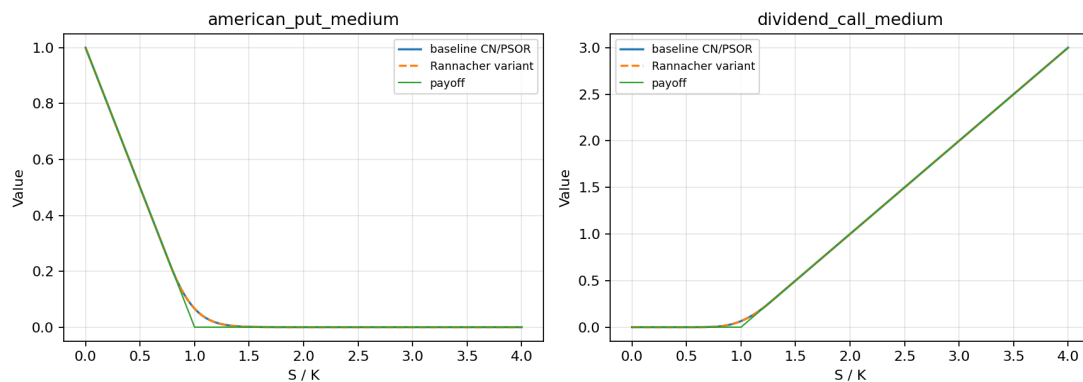


Figure 1: Baseline and Rannacher final value profiles nearly overlap for both cases.

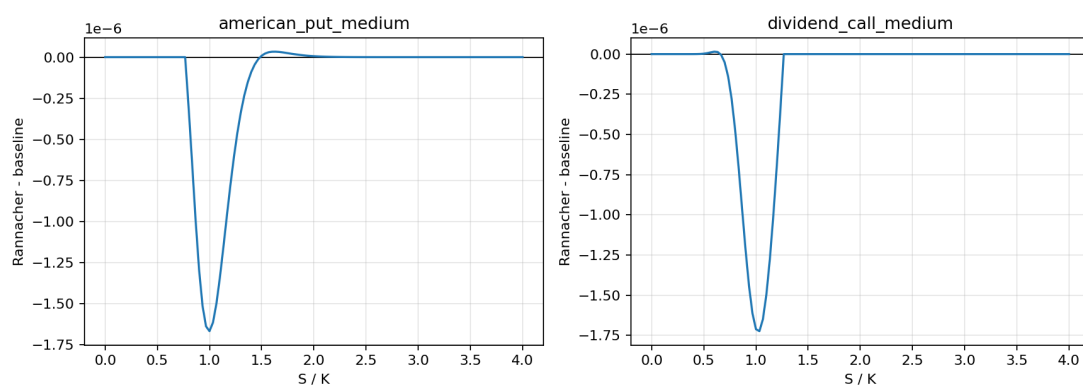


Figure 2: Smoothed-minus-baseline price differences are very small across moneyness.

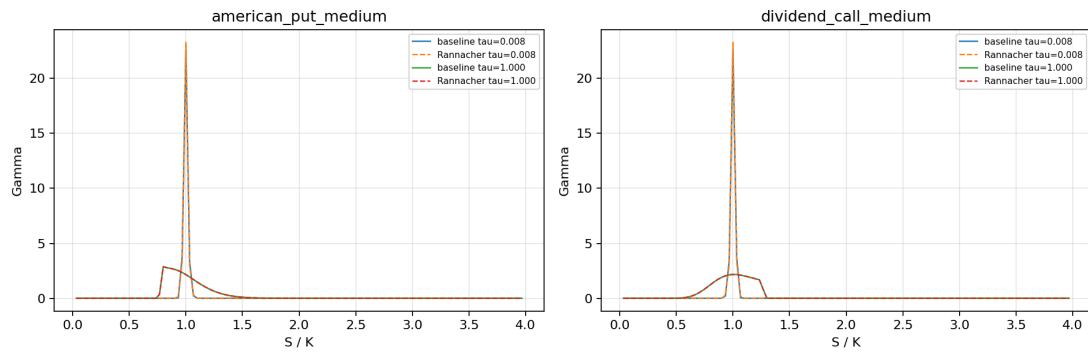


Figure 3: Gamma profiles show why the comparison is a diagnostic gate rather than a visual smoothing exercise.

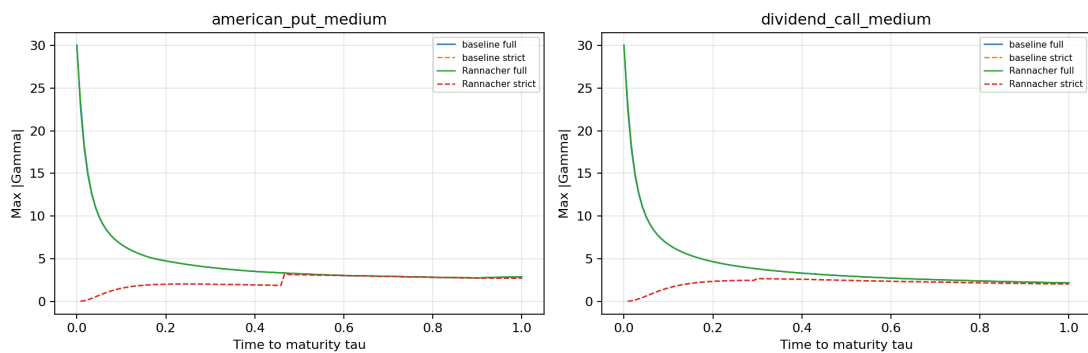


Figure 4: Full-grid and strict-mask max $|\Gamma|$ over time for baseline and Rannacher runs.

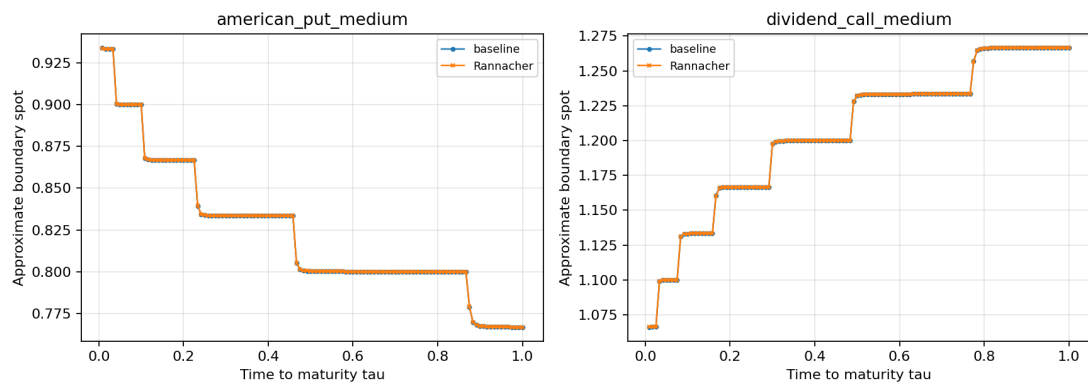


Figure 5: Threshold-based boundary curves are close, but remain approximate diagnostics.

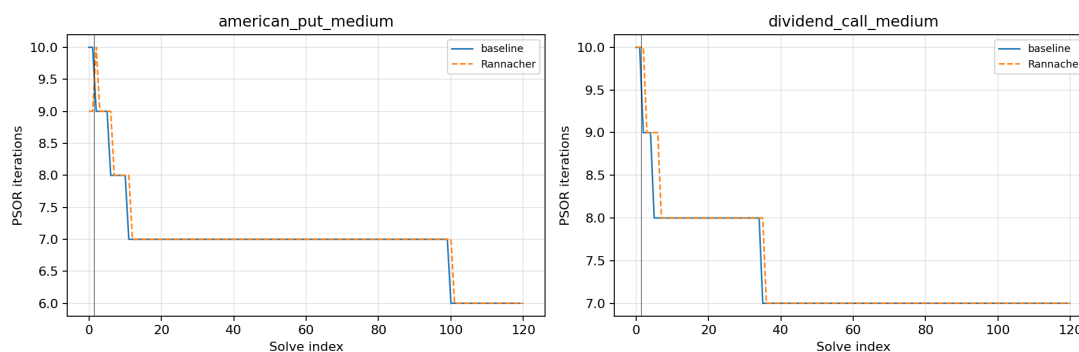


Figure 6: PSOR iteration counts remain well behaved for baseline and Rannacher solves.

14 Testing Strategy

Test category	Interpretation
Backward-Euler PSOR step	Checks shape, obstacle projection, boundary placement, and convergence metadata.
Rannacher result shape	Verifies that the smoothed result preserves the baseline grid and value-grid shape.
Zero substeps	Confirms that <code>rannacher_substeps=0</code> delegates to baseline behavior.
Obstacle checks	Confirms smoothed put and dividend-call values remain above payoff.
Price closeness	Checks that smoothed and baseline prices remain finite and close on representative cases.
Boundary and Greek diagnostics	Verifies that Ticket 09 and Ticket 10 diagnostics run on both result types.
Artifact checks	Confirms CSV columns and figure files are created.
Scope checks	Rejects stress-map, dataset, neural-network, or machine-learning-label scope creep.

15 Test Results and Interpretation

The final integrated command was:

```
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src MPLCONFIGDIR=/private/
tmp/matplotlib-cache FC_CACHEDIR=/private/tmp/fontconfig-cache python3 -m
unittest discover -s tests
```

Result:

```
Ran 135 tests in 2.901s
OK
```

The Ticket 10A-focused test file also passed:

```
Ran 14 tests in 0.895s
```

OK

These tests show that the named Rannacher variant, comparison experiment, CSV artifacts, and figures are reproducible in the current environment. They do not prove that Rannacher smoothing is globally better. That would require grid/domain sensitivity and broader parameter testing.

16 Implementation Pitfalls and Debugging Notes

Actual recorded implementation iterations:

Event	Symptom	Fix	Beginner lesson
Initial TDD red run	Full suite reported 9 errors because <code>rannacher.py</code> and the Ticket 10A experiment did not exist yet.	Implemented the new module and experiment files.	Red tests are useful when they fail for the expected missing capability.
Focused module invocation	<code>python-munittesttests.test_rannacher_smoothing...</code> failed because <code>tests/</code> is not a package.	Used unittest discovery with <code>-stests-ptest_rannacher_smoothing.py</code> .	Test invocation style must match repository structure.

Likely traps to watch for in future Rannacher work:

Pitfall	Why it matters
Silently changing the baseline solver	It would invalidate earlier validation tickets and hide the comparison.
Claiming smoothing is better because plots look smoother	The gate must check prices, boundaries, Greeks, and convergence.
Ignoring price shifts	A smoother Gamma is not acceptable if prices move materially.
Ignoring boundary shifts	Boundary curves are approximate, but large shifts are still warning signs.
Comparing different grids or parameters	The comparison would mix smoothing effects with discretization effects.
Using smoothed Greeks as dataset labels too early	Dataset and neural-surface work is out of scope and requires later validation.

17 Gate Conclusion with Evidence

The gate conclusion for Ticket 10A is:

KEEP_BASELINE_FOR_NOW

Evidence:

- price differences are very small, around 1.7×10^{-6} ;

- boundary shifts are small, below 7×10^{-4} ;
- PSOR convergence remains clean;
- strict-mask Gamma does not improve in either representative case.

Therefore, the Rannacher variant is useful as a named comparison tool, but the project should not adopt it as the default or claim that it improves Greek diagnostics yet. A later grid/domain sensitivity ticket can revisit this decision with a broader evidence base.

18 Limitations

Limitation	Explanation
Only two representative cases	The comparison is not full grid/domain sensitivity.
Only one smoothing setting	The default uses two half-substeps; other choices are not explored.
Boundary curves are threshold-based	Boundary shifts inherit Ticket 09 extraction uncertainty.
Greek masks are diagnostic	Strict-mask Gamma is useful, but not an analytical proof.
No stress maps or datasets	Ticket 10A intentionally avoids later machine-learning or stress-surface work.

19 Lessons Learned / Study Notes

For beginner researchers, the main lesson is that numerical smoothing is a trade-off. It may damp payoff-kink artifacts, but it is not automatically better. A careful comparison asks:

- Did prices stay close?
- Did obstacle and PSOR convergence remain clean?
- Did approximate boundaries shift?
- Did Gamma actually improve under the chosen diagnostic mask?

In this ticket, prices and boundaries stayed close, but Gamma did not improve. That is a valid and useful result.

20 Link to the Next Ticket

The next ticket should continue the validation ladder without treating Ticket 10A as a default solver change. A natural next step is grid/domain sensitivity: compare solver outputs across mesh sizes and domain cutoffs before using Greeks, boundaries, or prices as downstream research features.

21 Reproducibility Record

Item	Record
New source file	src/american_risk_surfaces/solvers/rannacher.py.
New experiment	experiments/08_rannacher_smoothing_comparison.py.
New test file	tests/test_rannacher_smoothing.py.
CSV outputs	Four files under results/01_solver_validation/tables/ with prefix ticket_10a.
Figure outputs	Six files under results/01_solver_validation/figures/ with prefix ticket_10a.
Report outputs	This L ^A T _E X source and compiled PDF under reports/03_solver/tickets/.